Projective Radial Isometry and Topological Foliation of Conservation Laws in Neural Networks

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Abstract

Deep neural networks face a structural tension between representation normalization and physical conservation laws. Standard normalizations (LayerNorm, RMSNorm) project activations onto a fixed hypersphere, destroying input norm (energy) information and collapsing Jacobian rank in low-dimensional bottlenecks, whereas linear unitary networks ($U^\top U = I$) cannot traverse dimension-changing bottlenecks ($d_{\text{out}} < d_{\text{in}}$) or model amplitude-dependent nonlinear frequencies. Here we introduce **SkipNorm**, $\hat{y} = \|x\|_p \, f(x)/\|f(x)\|_p$, and its tangent-space retraction **Geodesic SkipNorm**, establishing a class of $\mathcal{O}(d)$ nonlinear projective radial isometries. First, we prove the *Orthogonal Jacobian Decomposition* and *Bottleneck Rank Theorem*: across compositions of arbitrary depth L and varying widths $d_0, \ldots, d_L$, SkipNorm preserves the exact radial singular value $\sigma_r \equiv 1$ ($\left\|J_{\text{total}}^\top u_L - u_0\right\|_2 = 0$ to machine precision 10^{-15}) and maintains full Jacobian rank d in width-d bottlenecks where RMSNorm drops to rank $d-1$ and LayerNorm drops to rank $d-2$ (collapsing $d = 2$ to two discrete points). Second, we prove the *Topological Noether-Brouwer Foliation Theorem*: when an autonomous dynamical system on $\mathcal{M} \subset \mathbb{R}^N$ ($N \geq 2$) with compact, connected $(N-1)$-dimensional regular level sets $\mathcal{O}_c = I^{-1}(c)$ is conjugated into a dimension-matched Geodesic SkipNorm latent space ($d_z = N$), Brouwer's Invariance of Domain Theorem forces each $\mathcal{O}_c$ to map homeomorphically onto a concentric sphere $\mathbb{S}^{N-1}(\psi(c))$ via a strictly monotonic homeomorphism $\psi(c)$ of the conservation law $I(x)$. Our theorem also predicts two exact topological obstructions—multi-basin bifurcation when $\mathcal{O}_c$ is disconnected (verified across a double-well homoclinic separatrix) and non-homeomorphism on higher-genus tori $\mathbb{T}^K \ncong \mathbb{S}^{2K-1}$ ($K \geq 2$), which we resolve via **Toroidal Block SkipNorm** on $\bigoplus_{k=1}^{K} \mathbb{R}^2$. In 64-bit JAX experiments, our framework conserves lattice wave norm to 10^{-13}% over 2,000 steps and discovers non-canonical Casimir and multi-action invariants with $R^2 = 94.6\%$–98.4% without prior symplectic structure or contrastive supervision.

Introduction.—Symmetry and conservation laws govern the evolution of physical systems through Noether's theorem and Liouville-Arnold integrability [1], yet incorporating exact conservation into deep neural networks remains an open challenge. In modern deep learning, normalization layers such as Layer Normalization (LayerNorm) [2] and Root Mean Square Normalization (RMSNorm) [3, 4] stabilize signal propagation by projecting intermediate representations $y = f(x) \in \mathbb{R}^{d_{\text{out}}}$ onto a manifold of constant norm $\|\hat{y}\|_2 = \sqrt{d_{\text{out}}}$. By construction, this projection annihilates the input norm $\|x\|_2$, discarding global signal energy, contrast, or physical amplitude at every layer. Conversely, equivariant radial gates such as ModReLU and $SO(3)$-equivariant activations [5, 6] update the norm while freezing the unit direction $x/\|x\|_2$:

$$\underbrace{\hat{y}_{\text{eq}} = g(\|x\|_2)\frac{x}{\|x\|_2}}_{\text{Preserves direction, maps norm}} \quad \leftrightarrow \quad \underbrace{\hat{y}_{\text{SN}} = \|x\|_p \, \frac{f(x)}{\|f(x)\|_p}}_{\text{Preserves norm, maps direction}} \quad . \tag{1}$$

As shown in Eq. (1), the operation $\hat{y} = \|x\|_p \, f(x)/\|f(x)\|_p$—which we term **SkipNorm**—is the exact *polar dual* of equivariant radial gating: a multiplicative skip connection that routes the L_p norm $\|x\|_p$ of the input around an arbitrary nonlinear transformation $f : \mathbb{R}^{d_{\text{in}}} \to \mathbb{R}^{d_{\text{out}}}$.

While linear unitary and orthogonal networks ($U^\top U = I$) preserve Euclidean norm [5, 7–9], they face two fundamental obstructions: (i) by the Rank-Nullity Theorem, no linear map can be norm-preserving when compressing dimensions ($d_{\text{out}} < d_{\text{in}}$), and (ii) a constant unitary matrix U has fixed state-independent eigenfrequencies, rendering it unable to represent nonlinear oscillators or dispersive waves whose phase velocity depends on amplitude. Meanwhile, in physics-informed machine learning, Hamiltonian Neural Networks (HNNs) [10], Lagrangian networks [11], SympNets [12], and Action-Angle networks [13, 14] require canonical symplectic coordinates (q, p) where the skew-symmetric Poisson tensor J is known *a priori*. Furthermore, by the Ge-Marsden theorem [15], discrete symplectic maps that preserve the phase-space 2-form $dq \wedge dp$ cannot exactly conserve a nonlinear Hamiltonian $H(q, p)$ unless they coincide with the exact time-reparameterized flow, accumulating truncation drift over long rollouts. Existing methods for automated conservation-law discovery from non-canonical trajectories—such as AI Poincaré [16, 17], Siamese invariant classifiers [18, 19], or post-hoc Hessian eigendecompositions [20, 21]—either diagnose in-

variant manifold dimension offline without providing a forward dynamical model, or require trajectory-segmented contrastive pairs.

In this Letter, we develop the differential geometry and algebraic topology of SkipNorm and its tangent-space extensions (**Geodesic SkipNorm** and **Toroidal Block SkipNorm**). We prove that SkipNorm acts as an exact projective radial isometry across arbitrary nonlinearities and rectangular bottlenecks, and we establish the *Topological Noether-Brouwer Foliation Theorem*: a dimension-matched Geodesic SkipNorm autoencoder trained solely on trajectory prediction topologically forces its latent spheres (or tori) to foliate the unknown conservation laws of an autonomous dynamical system, while predicting exact topological obstructions across multi-basin separatrices and higher-genus tori.

Projective radial isometry and bottleneck rank.—Let $x \in \mathbb{R}^{d_{\text{in}}} \setminus \{0\}$ and let $f : \mathbb{R}^{d_{\text{in}}} \to \mathbb{R}^{d_{\text{out}}}$ be a C^1 mapping such that $y = f(x) \neq 0$. For $p \in [1, \infty]$, the SkipNorm operator is defined as

$$\text{SN}_p(x; f) := \frac{f(x)}{\max(\|f(x)\|_p, \epsilon)} \|x\|_p , \tag{2}$$

where $\epsilon \approx 10^{-30}$ in `float64` prevents division by zero at the origin without introducing the cumulative $1/(1 + \epsilon/\|y\|_p)$ damping artifact of additive denominators. Geometrically, under the polar diffeomorphism $\mathbb{R}^d \setminus \{0\} \cong \mathbb{S}_p^{d-1} \times \mathbb{R}_{>0}$, SkipNorm is a fiber-bundle morphism over the radial base space $\mathbb{R}_{>0}$ that acts as a nonlinear map on the directional fiber $\mathbb{S}_p^{d-1}$ while acting as the exact identity on the radial coordinate $r = \|x\|_p$.

Our first theorem characterizes the exact Jacobian spectrum and deep gradient propagation of SkipNorm for $p = 2$ (complete proofs of all results are provided in the Supplemental Material).

Theorem 1 (Orthogonal Jacobian Decomposition & Deep Radial Isometry)**.** *Let $\hat{y} = \text{SN}_2(x; f)$ with $x \in \mathbb{R}^{d_{\text{in}}} \setminus \{0\}$, $y = f(x) \in \mathbb{R}^{d_{\text{out}}} \setminus \{0\}$, and unit radial vectors $u_x = x/\|x\|_2$, $u_y = y/\|y\|_2 = \hat{y}/\|\hat{y}\|_2$. Let $J_f = \partial y/\partial x \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$. The Jacobian $J = \partial \hat{y}/\partial x$ decomposes into two mutually orthogonal operators:*

$$J = \underbrace{\frac{\|x\|_2}{\|y\|_2} \left(I_{d_{\text{out}}} - u_y u_y^\top\right) J_f}_{A \text{ (Tangential: } u_y^\top A = 0)} + \underbrace{u_y u_x^\top}_{B \text{ (Radial: rank 1)}} . \tag{3}$$

Consequently, for any composition of L SkipNorm layers of arbitrary varying widths $d_0 \to d_1 \to \cdots \to d_L$ with total Jacobian $J_{\text{total}} = J_L J_{L-1} \cdots J_1$:

$$J_{\text{total}}^\top u_L \equiv u_0, \qquad \left\|J_{\text{total}}^\top u_L\right\|_2 \equiv 1. \tag{4}$$

Moreover, if each f_l is positively 1-homogeneous ($f_l(cx) = cf_l(x)$ for $c > 0$, e.g., bias-free LeakyReLU *or* ELU *networks), then $J_f u_x = (\|y\|_2 / \|x\|_2) u_y \in \ker(I - u_y u_y^\top)$, so $Au_x = 0$ and J_{total} is a strict left- and right-singular radial isometry ($J_{\text{total}} u_0 = u_L$, $\sigma_{\text{radial}} = 1$).*

Equation (4) reveals a remarkable property: even when a deep network passes through narrow bottlenecks ($d_l \ll d_0$) where linear orthogonal matrices cannot exist, the radial gradient $J_{\text{total}}^\top u_L$ is transported backward across arbitrary depth L with *zero attenuation or amplification*, recovering the exact input direction u_0 [Fig. 1(b)]. Theorem 1 also explains why standard normalizations suffer catastrophic representational collapse inside low-dimensional bottlenecks.

Theorem 2 (Bottleneck Rank Theorem)**.** *Consider a representation layer of width $d \geq 2$ at any non-degenerate point $x \in \mathbb{R}^d \setminus \{0\}$ where $\text{rank}(J_f(x)) = d$ and $J_f(x) u_x \notin \text{span}(u_y)^\perp$:*

1. ***LayerNorm*** *($\hat{y}_{\text{LN}} = \sqrt{d}\, P_{\perp \mathbf{1}} y / \|P_{\perp \mathbf{1}} y\|_2$)**:** Both the mean direction $\mathbf{1}$ and the centered radial direction u_{y_c} lie in the left nullspace of J_{LN}, so $\text{rank}(J_{\text{LN}}) \leq d - 2$. At $d = 2$, $\text{rank}(J_{\text{LN}}) = 0$ everywhere and $\mathbb{R}^2$ collapses onto two discrete points $\{\pm(1, -1)\}$.*

2. ***RMSNorm / SphericalNorm*** *($\hat{y}_{\text{RMS}} = \sqrt{d}\, y / \|y\|_2$)**:** The radial direction u_y lies in the left nullspace of J_{RMS}, so $\text{rank}(J_{\text{RMS}}) \leq d - 1$ (collapsing $d = 2$ to a* 1D *circle).*

3. ***SkipNorm*** *($\hat{y}_{\text{SN}} = \|x\|_2\, y / \|y\|_2$)**:** If $P_{\perp u_y} J_f$ has rank $d - 1$ on $T_x \mathbb{S}^{d-1}$ (satisfied generically by smooth or bi-orthant activations), then $\text{rank}(J_{\text{SN}}) = d$ (full rank).*

Theorem 2 is verified to machine precision in Fig. 1(a,c). At $d = 2$ in `float64`, the minimum singular value of LayerNorm is 6.4×10^{-34} (rank = 0), RMSNorm is 9.9×10^{-17} (rank = 1), and SkipNorm is $\sigma_{\min} = 0.264$ (rank = 2). Theorem 1 also implies an important activation rule: placing a strict ReLU immediately inside SkipNorm traps outputs in the positive orthant $\mathbb{S}_+^{d-1} = \mathbb{S}^{d-1} \cap \mathbb{R}_{\geq 0}^d$; in a $d = 2$ bottleneck, whenever only one neuron is active, $\hat{y}$ is pinned to a coordinate axis with zero tangential gradient. Using smooth or bi-orthant activations (SiLU, GELU, LeakyReLU) preserves the full sphere $\mathbb{S}^{d-1}$ and full rank d.

Theorem 3 (Scale Equivariance & Weight 0-Homogeneity)**.** *If $f(x; W) = \phi(Wx)$ where ϕ is positively 1-homogeneous, then $\text{SN}_p(cx; \beta W) = c\, \text{SN}_p(x; W)$ for all $c, \beta > 0$, and $\langle \nabla_W \text{SN}_p(x; W), W \rangle_F = 0$.*

Tangent-space geodesic retraction and nonlinear lattice waves.—When modeling continuous-time or recurrent dynamical systems $x_{t+1} = \Phi_{\Delta t}(x_t)$, applying naive SkipNorm $\hat{y} = \|x\|_2 f(x)/\|f(x)\|_2$ allows $f(x)$ to point in an arbitrary direction unrelated to x, while an additive residual $y = x + f(x)$ suffers an antipodal phase-flip singularity if $f(x) \approx -2x$. To construct a smooth near-identity flow on concentric spheres $\mathbb{S}^{d-1}(\|x\|_2)$, we project the residual update onto the tangent hyperplane $T_x \mathbb{S}^{d-1} = \{v \in \mathbb{R}^d : \langle v, x \rangle = 0\}$ prior to radial retraction,

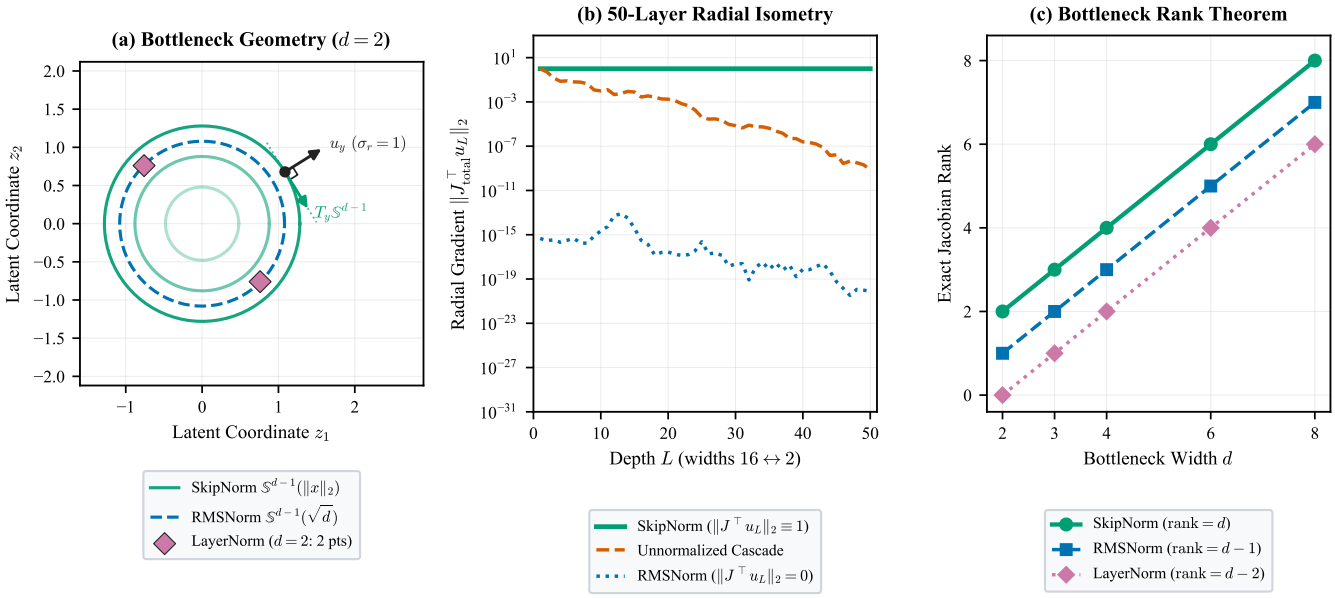


Figure 1: Differential Geometry and Bottleneck Rank of SkipNorm. **(a)** In a $d = 2$ bottleneck, LayerNorm collapses all of $\mathbb{R}^2$ onto two discrete points (rank $= 0$), and RMSNorm collapses $\mathbb{R}^2$ onto a single circle $\mathbb{S}^1(\sqrt{2})$ (rank $= 1$). SkipNorm preserves the full concentric foliation $\mathbb{S}^{d-1}(\|x\|_2) \times \mathbb{R}_{>0}$ (rank $= 2$) with unit radial singular value $\sigma_r = 1$ along u_y and unconstrained angular flexibility along the orthogonal tangent space $T_y\mathbb{S}^{d-1}$. **(b)** Across a 50-layer deep bottleneck cascade with widths oscillating between 16 and 2 in 64-bit precision (`float64`), SkipNorm maintains exact radial gradient isometry $\left\|J_{\text{total}}^\top u_L\right\|_2 = 1.0 \pm 1.1 \times 10^{-15}$, whereas unnormalized cascades decay exponentially and RMSNorm annihilates radial gradients ($\sim 10^{-17}$). **(c)** Exact Jacobian rank across bottleneck widths $d \in \{2, 3, 4, 6, 8\}$, verifying Theorem 2 ($\text{rank}_{\text{LN}} = d - 2$, $\text{rank}_{\text{RMS}} = d - 1$, $\text{rank}_{\text{SN}} = d$).

defining **Geodesic SkipNorm**:

$$f_\perp(x) = f(x) - \frac{\langle f(x), x\rangle}{\max(\|x\|_2^2, \epsilon^2)} x,$$
$$\text{GSN}(x; f, \alpha) = \frac{x + \alpha f_\perp(x)}{\sqrt{\|x\|_2^2 + \alpha^2 \|f_\perp(x)\|_2^2}} \|x\|_2, \qquad (5)$$

where $\alpha \in \mathbb{R}$ is a learnable step scale. Because $f_\perp(x) \perp x$, the denominator $\sqrt{\|x\|_2^2 + \alpha^2 \|f_\perp(x)\|_2^2} \geq \|x\|_2$ is strictly bounded away from zero for all $\alpha \in \mathbb{R}$, and $\langle \text{GSN}(x), x\rangle > 0$, eliminating antipodal flips and yielding a first-order retraction onto $\mathbb{S}^{d-1}(\|x\|_2)$ in $\mathcal{O}(d)$ time.

To test GeodesicSkipNorm on a high-dimensional quantum/optical wave system where linear unitary networks fail, we consider the 1D Discrete Nonlinear Schrödinger (DNLS) equation on a periodic lattice of $L = 16$ complex sites ($\psi \in \mathbb{C}^{16} \cong \mathbb{R}^{32}$):

$$i\frac{d\psi_n}{dt} = -(\psi_{n+1} + \psi_{n-1} - 2\psi_n) + g\,|\psi_n|^2\,\psi_n, \qquad (6)$$

with Kerr nonlinearity $g = 1.5$. Equation (6) conserves the total wave norm $\mathcal{N}(\psi) = \sum_{n=1}^{L} |\psi_n|^2 = \|\psi\|_2^2$, while its local phase rotation rate depends nonlinearly on intensity $|\psi_n|^2$. As shown in Fig. 2 and Table 1, across 5 random seeds, a standard ResNet stepper diverges to $7.2 \times 10^{71}\%$ norm error over 2,000 steps, and RMSNorm destroys the initial amplitude ($1.2 \times 10^3\%$ error). A linear unitary operator ($\hat{\psi} = U\psi$ via the Cayley transform $U = (I - A)(I + A)^{-1}$, $A^\top = -A$) preserves norm but incurs 2.08× higher trajectory MSE (0.2079 vs. 0.1001) because a constant matrix U cannot capture the state-dependent frequency $g\,|\psi_n|^2$. GeodesicSkipNorm achieves the lowest trajectory error while conserving $\mathcal{N}(\psi)$ to $7.38 \times 10^{-13}\% \pm 1.30 \times 10^{-13}\%$.

Topological Noether-Brouwer foliation of conservation laws.—In general physical and biological systems, conserved quantities are not the raw input Euclidean norm $\|x\|_2$; rather, they are nonlinear functions $I(x)$ (such as energy, Casimir invariants, or action integrals) defined in non-canonical coordinates where $\|x\|_2$ oscillates wildly along each orbit. Suppose we train a latent autoencoder $(\Phi_{\text{enc}}, \Phi_{\text{dec}})$ with $\Phi_{\text{enc}} : \mathcal{M} \subset \mathbb{R}^N \to \mathbb{R}^{d_z}$ and a Geodesic SkipNorm latent stepper $g_\theta : \mathbb{R}^{d_z} \to \mathbb{R}^{d_z}$ ($\|g_\theta(z)\|_2 \equiv \|z\|_2$) to predict trajectories $x_{t+k} = \Phi_{\text{dec}}(g_\theta^{(k)}(\Phi_{\text{enc}}(x_t)))$. Because g_θ preserves the latent norm $\|z\|_2$ at every step, intertwining the true flow $\Phi_{\text{flow}}^{\Delta t}$ with g_θ forces $\|\Phi_{\text{enc}}(x)\|_2$ to remain constant along each trajectory.

Crucially, however, orbital constancy alone does not

Table 1: 1D Discrete Nonlinear Schrödinger ($L = 16, \mathbb{R}^{32}$) benchmark across 5 random seeds (mean ± SD in `float64`).

Architecture	40-Step Rollout MSE	2,000-Step $\Delta\mathcal{N}/\mathcal{N}_0$ (%)
Standard ResNet	0.1026 ± 0.0108	$7.15 \times 10^{71} \pm 1.43 \times 10^{72}$
RMSNorm	0.7206 ± 0.0223	$1.20 \times 10^{3} \pm 1.76 \times 10^{-13}$
Linear Unitary ($U^\top U = I$)	0.2079 ± 0.0000	$1.32 \times 10^{-11} \pm 1.14 \times 10^{-11}$
GeodesicSkipNorm	$\mathbf{0.1001 \pm 0.0056}$	$\mathbf{7.38 \times 10^{-13} \pm 1.30 \times 10^{-13}}$

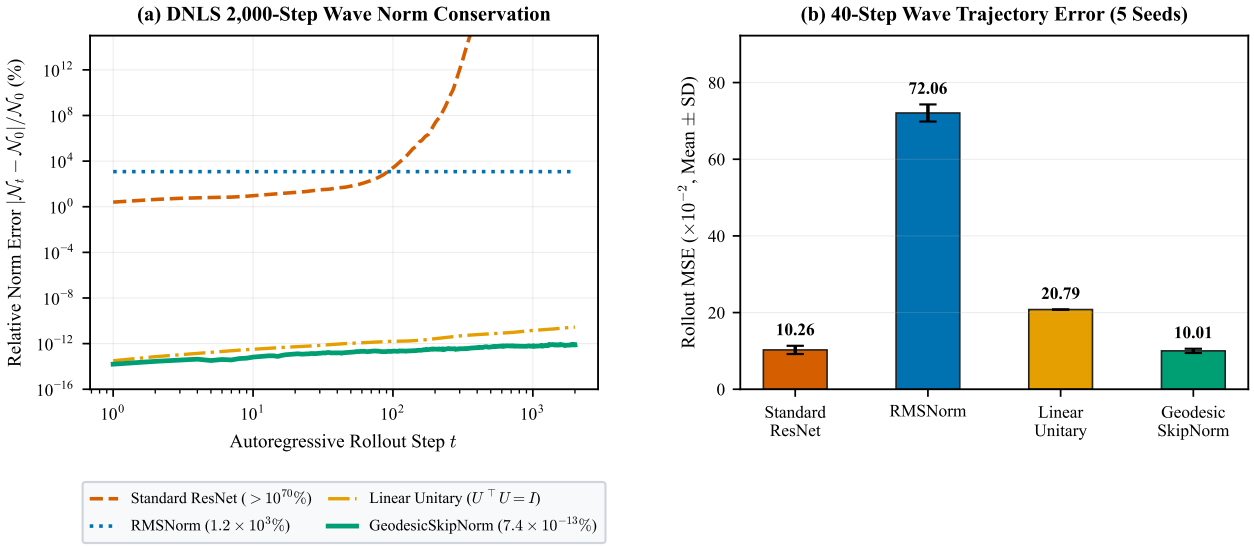


Figure 2: Unitary Lattice Wave Propagation in the 1D Discrete Nonlinear Schrödinger (DNLS) Equation. (a) Relative wave norm error $|\mathcal{N}_t - \mathcal{N}_0|/\mathcal{N}_0$ ($\mathcal{N} = \sum_{n=1}^{16} |\psi_n|^2$) across a 2,000-step autoregressive rollout in `float64`. Standard ResNet diverges catastrophically ($> 10^{70}$%), and RMSNorm destroys the initial wave amplitude (1.2×10^3% error), whereas GeodesicSkipNorm conserves wave norm to 7.38×10^{-13}%. **(b)** 40-step wave trajectory MSE across 5 random seeds (mean ± SD). GeodesicSkipNorm achieves 2.08× lower error than a strictly linear unitary operator ($U^\top U = I$), which cannot represent the amplitude-dependent nonlinear phase speed $g\,|\psi_n|^2$.

guarantee that $\|\Phi_{\text{enc}}(x)\|_2$ recovers the conservation law $I(x)$: if the latent dimension is overparameterized ($d_z > N$), a sphere $\mathbb{S}^{d_z-1}(r_0)$ has dimension $d_z - 1 \geq N$, so an injective encoder can embed the entire N-dimensional phase space $\mathcal{M}$ into a *single* sphere of constant radius r_0, yielding a trivial constant $\|\Phi_{\text{enc}}(x)\|_2 \equiv r_0$ across all orbits. Our central topological theorem proves that matching the latent bottleneck dimension to the state manifold dimension ($d_z = N$) eliminates this degeneracy via Brouwer's Invariance of Domain [24], while also revealing the exact topological conditions (compactness, connectedness, and spherical homeomorphism type) required for a global foliation.

Theorem 4 (Topological Noether-Brouwer Foliation Theorem)**.** *Let $\mathcal{M} \subset \mathbb{R}^N$ ($N \geq 2$) be an open domain around an equilibrium $x^* \in \mathcal{M}$, and let $I : \mathcal{M} \to \mathbb{R}$ be a C^1 conservation law whose regular values $C_{\text{reg}} = (c_{\min}, c_{\max})$ ($\nabla I(x) \neq 0$ on $I^{-1}(C_{\text{reg}})$) foliate $\mathcal{M} \setminus \{x^*\}$ into **compact, connected** $(N-1)$-dimensional level sets $\mathcal{O}_c = I^{-1}(c)$ shrinking to $\mathcal{O}_{c_{\min}} = \{x^*\}$ as $c \downarrow c_{\min}$. Suppose the discrete flow $\Phi_{\text{flow}}^{\Delta t}$ has a dense orbit on each $\mathcal{O}_c$. If a continuous, **globally injective** encoder $\Phi_{\text{enc}} : \mathcal{M} \to \mathbb{R}^N$ (into the matched dimension $d_z = N$) intertwines the flow with a SkipNorm operator $g_\theta : \mathbb{R}^N \to \mathbb{R}^N$ ($\Phi_{\text{enc}} \circ \Phi_{\text{flow}}^{\Delta t} = g_\theta \circ \Phi_{\text{enc}}$), then:*

1. ***Spherical Homeomorphism:*** *There exists a continuous function $\psi : (c_{\min}, c_{\max}) \to \mathbb{R}_{>0}$ such that for every $c \in (c_{\min}, c_{\max})$, the restriction $\Phi_{\text{enc}}|_{\mathcal{O}_c} : \mathcal{O}_c \to \mathbb{S}^{N-1}(\psi(c))$ is a **homeomorphism** onto the entire $(N-1)$-sphere:*

$$\Phi_{\text{enc}}(\mathcal{O}_c) = \mathbb{S}^{N-1}(\psi(c)), \qquad \mathcal{O}_c \cong \mathbb{S}^{N-1}. \tag{7}$$

2. ***Strict Monotonicity & Origin Alignment:*** *The map $\psi : (c_{\min}, c_{\max}) \to (r_{\min}, r_{\max}) \subseteq \mathbb{R}_{>0}$ is a strictly monotonic homeomorphism onto an open image interval $(r_{\min}, r_{\max})$ ($\psi(c_1) \neq \psi(c_2)$ for $c_1 \neq c_2$), with $\lim_{c \downarrow c_{\min}} \psi(c) = 0$ and $\Phi_{\text{enc}}(x^*) = 0$.*

Proof Sketch. By exact intertwining and radial isometry $\|g_\theta(z)\|_2 = \|z\|_2$, the continuous map $h(x) := \|\Phi_{\text{enc}}(x)\|_2$ is invariant along every discrete orbit $\{\Phi_{\text{flow}}^{k\Delta t}(x_0)\}_{k \in \mathbb{N}}$. By orbit density on $\mathcal{O}_c$, h is constant on each connected level set $\mathcal{O}_c$, defining $h|_{\mathcal{O}_c} \equiv \psi(c)$. Since I is a C^1 submersion on $I^{-1}(C_{\text{reg}})$, the Submersion Theorem guarantees continuous local sections $s : U \to \mathcal{M}$ ($I(s(c)) = c$), so $\psi(c) = h(s(c))$ is continuous. Because $N \geq 2$ and $\mathcal{O}_c$ is a connected $(N-1)$-manifold containing multiple points, global injectivity of Φ_{enc} implies $\psi(c) > 0$. The restriction $\Phi_{\text{enc}}|_{\mathcal{O}_c} : \mathcal{O}_c \to \mathbb{S}^{N-1}(\psi(c))$ is a continuous injection between $(N-1)$-dimensional topological manifolds without boundary. By Brouwer's Invariance of Domain Theorem, $\Phi_{\text{enc}}(\mathcal{O}_c)$ is open in $\mathbb{S}^{N-1}(\psi(c))$; since $\mathcal{O}_c$ is compact and $\mathbb{S}^{N-1}(\psi(c))$ is Hausdorff, $\Phi_{\text{enc}}(\mathcal{O}_c)$ is also closed. Since $\mathbb{S}^{N-1}(\psi(c))$ is connected for $N \geq 2$, $\Phi_{\text{enc}}(\mathcal{O}_c) = \mathbb{S}^{N-1}(\psi(c))$, and as a continuous bijection from a compact space to a Hausdorff space, $\Phi_{\text{enc}}|_{\mathcal{O}_c}$ is a homeomorphism. Finally, for $c_1 \neq c_2$, $\mathcal{O}_{c_1} \cap \mathcal{O}_{c_2} = \emptyset$, so global injectivity of Φ_{enc} forces $\mathbb{S}^{N-1}(\psi(c_1)) \cap \mathbb{S}^{N-1}(\psi(c_2)) = \emptyset$, yielding $\psi(c_1) \neq \psi(c_2)$. A continuous injective function on an interval $(c_{\min}, c_{\max})$ is a strictly monotonic homeomorphism onto its open image interval $(r_{\min}, r_{\max})$, and $\text{diam}(\mathcal{O}_c) \to 0$ as $c \downarrow c_{\min}$ forces $\Phi_{\text{enc}}(x^*) = 0$. □

Each hypothesis of Theorem 4 is sharp, and relaxing either connectedness of $\mathcal{O}_c$ or spherical topology yields

two physical corollaries that govern multi-basin potentials and integrable K-tori.

Corollary 1 (Multi-Basin Obstruction for Disconnected $\mathcal{O}_c$)**.** *Suppose a regular level set $\mathcal{O}_c = \bigsqcup_{m=1}^{M} \mathcal{O}_c^{(m)}$ has $M \geq 2$ disjoint compact connected components in $\mathbb{R}^N$ ($N \geq 2$, e.g., a double-well Hamiltonian below a saddle separatrix). Then **no** continuous globally injective encoder $\Phi_{\text{enc}} : \mathcal{M} \to \mathbb{R}^N$ can have constant norm $\|\Phi_{\text{enc}}(x)\|_2 = \psi(c)$ on all of $\mathcal{O}_c$: by Brouwer's Invariance of Domain applied to each component, $\|\Phi_{\text{enc}}\|_2 \equiv r_m$ on $\mathcal{O}_c^{(m)}$ forces $\Phi_{\text{enc}}(\mathcal{O}_c^{(m)}) = \mathbb{S}^{N-1}(r_m)$, so $r_1 = r_2$ would force two disjoint components to cover the same sphere, violating injectivity. Moreover, since distinct wells bound disjoint topological disks $D^{(1)} \cap D^{(2)} = \emptyset$, at most one well equilibrium can map to $0 \in \mathbb{R}^N$.*

Corollary 2 (Toroidal Obstruction for $\mathbb{T}^K \not\cong \mathbb{S}^{2K-1}$)**.** *Because Eq. (7) requires $\mathcal{O}_c \cong \mathbb{S}^{N-1}$, no continuous injective single-sphere SkipNorm encoder into $\mathbb{R}^{2K}$ can homeomorphically foliate an integrable system with $K \geq 2$ commuting invariants whose invariant manifolds are Liouville-Arnold K-tori $\mathbb{T}^K \cong (\mathbb{S}^1)^K \not\cong \mathbb{S}^{2K-1}$.*

To resolve the toroidal obstruction of Corollary 2, we partition the latent space into K two-dimensional planes $z = (z^{(1)}, \ldots, z^{(K)}) \in \bigoplus_{k=1}^{K} \mathbb{R}^2 \cong \mathbb{R}^{2K}$ and apply Geodesic SkipNorm independently to each block's radius while allowing $f(z)$ to couple all blocks nonlinearly, defining **Toroidal Block SkipNorm**:

$$\hat{z}^{(k)} = \text{GSN}\big(z^{(k)};\, f^{(k)}(z^{(1)}, \ldots, z^{(K)}),\, \alpha_k\big),\ k = 1, \ldots, K. \tag{8}$$

Equation (8) preserves all K block radii $r_k = \big\|z^{(k)}\big\|_2$, foliating $\mathbb{R}^{2K}$ into $\prod_{k=1}^{K} \mathbb{S}^1(r_k) \cong \mathbb{T}^K$—constructing coordinate-free neural Liouville-Arnold action-angle variables where $r_k = \sqrt{2J_k}$ are the conserved actions and $z^{(k)}/r_k \in \mathbb{S}^1$ are the coupled nonlinear angles.

Finally, while Theorem 4 analyzes an idealized exact intertwining ($\Phi_{\text{enc}} \circ \Phi_{\text{flow}}^{\Delta t} = g_\theta \circ \Phi_{\text{enc}}$), practical neural networks are trained to a finite commutation error $\varepsilon > 0$. The following stability bound explains why trained networks achieve empirical $R^2 \approx 95\%$–99.9%.

Proposition 1 (Finite-Error Approximate Conjugacy Bound)**.** *Suppose an L_{enc}-Lipschitz encoder Φ_{enc} and SkipNorm stepper g_θ ($\|g_\theta(z)\|_2 = \|z\|_2$) satisfy $\sup_{x \in \mathcal{M}} \big\|\Phi_{\text{enc}}(\Phi_{\text{flow}}^{\Delta t}(x)) - g_\theta(\Phi_{\text{enc}}(x))\big\|_2 \leq \varepsilon$, and suppose an M-step trajectory $\{x_k = \Phi_{\text{flow}}^{k\Delta t}(x_0)\}_{k=0}^{M}$ is δ_M-dense in $\mathcal{O}_c$. Then the intra-orbit radius oscillation on $\mathcal{O}_c$ is bounded by*

$$\sup_{x, x' \in \mathcal{O}_c} \big|\|\Phi_{\text{enc}}(x)\|_2 - \|\Phi_{\text{enc}}(x')\|_2\big| \leq M\varepsilon + 2L_{\text{enc}}\delta_M. \tag{9}$$

Discovering non-canonical invariants and verifying topological obstructions.—We now test Theorem 4 and Corollaries 1–2 across three dynamical systems (5 random seeds each in `float64`; Table 2). First, while origin-centered systems such as the quartic Duffing oscillator $H(q,p) = \frac{1}{2}p^2 + \frac{1}{4}q^2 + \frac{1}{2}q^4$ yield $R^2 = 99.87\%$ and $|\rho| = 1.000$ (Supplemental Material), a stringent test of Theorem 4 requires a non-canonical system whose true invariant $I(x)$ is completely uncorrelated with $\|x\|_2$. We consider the Lotka-Volterra predator-prey equations in the positive quadrant $(u, v) \in \mathbb{R}^2_{>0}$:

$$\dot{u} = \frac{2}{3}u - \frac{4}{3}uv, \qquad \dot{v} = uv - v, \tag{10}$$

whose equilibrium lies at $(u^*, v^*) = (1.0, 0.5) \neq (0, 0)$ and whose trajectories conserve the asymmetric logarithmic Casimir $V(u,v) = u - \ln u + \frac{4}{3}v - \frac{2}{3}\ln v$. Along a single orbit, the raw Euclidean norm $\|(u,v)\|_2$ varies by a factor of 7 [CV = 42.33%; Fig. 3(b)], yielding virtually zero Noether invariance ($R^2 = 1 - \mathbb{E}[\text{Var}_t(\|z\|_2)]/\text{Var}(\|z\|_2) = 0.09\%$). Training autoencoders over a 6-step horizon ($T_{\text{roll}} = 1.5\,\text{s}$) with zero prior knowledge of $V(u,v)$ or (u^*, v^*), standard ResNet ($d_z = 2$) and RMSNorm ($d_z = 2$) fail completely ($R^2 = 1.04\%$ and 0.96%). Overparameterized SkipNorm ($d_z = 8 > N = 2$) achieves small within-orbit variation (CV = 1.97%) by embedding the 2D plane into a nearly constant-radius shell in $\mathbb{S}^7$, yielding poor invariant recovery ($R^2 = 28.00\% \pm 21.15\%$). When dimension-matched ($d_z = N = 2$), GeodesicSkipNorm spontaneously maps $(1.0, 0.5) \mapsto (0, 0)$ and discovers $V(u,v)$ with $R^2 = 94.58\% \pm 1.84\%$ and Spearman correlation $|\rho| = 0.9744 \pm 0.0077$ [Fig. 3(a–c)].

Second, to test the multi-basin topological obstruction of Corollary 1, we train a $d_z = 2$ GeodesicSkipNorm autoencoder on the Double-Well Hamiltonian $H(q,p) = \frac{1}{2}p^2 - \frac{1}{2}q^2 + \frac{1}{4}q^4$, which possesses two symmetric potential wells centered at $(q^*, p^*) = (\pm 1, 0)$ separated by a figure-eight homoclinic saddle separatrix at $H = 0$ [Fig. 3(d)]. Exactly as predicted by Corollary 1, a sharp topological transition occurs at $H = 0$ [Fig. 3(e) and Table 2]: below the barrier ($H \in (-0.25, 0)$), where $\mathcal{O}_c = \mathcal{O}_c^{(L)} \sqcup \mathcal{O}_c^{(R)}$ has two disconnected components bounding disjoint disks, the injective encoder cannot center both wells at $0 \in \mathbb{R}^2$, resulting in high radial oscillation (CV = $24.89\% \pm 1.02\%$, $R^2 = 39.70\% \pm 2.08\%$). Immediately above the barrier ($H > 0$), where $\mathcal{O}_c$ merges into a single connected loop enclosing both wells, Theorem 4 takes effect and the exact same network achieves CV = $2.84\% \pm 0.44\%$,

Table 2: Discovery of conservation laws and verification of topological obstructions across 5 random seeds (mean ± SD in `float64`).

System / Regime / Architecture	Orbit CV (%)	Invariance R^2 (%)	Spearman $\|\rho\|$
1. Non-Centered Lotka-Volterra Casimir $V(u,v)$ ($N = 2$, Connected $\mathcal{O}_c$):			
Raw Input $\|(u,v)\|_2$	42.33	0.09	0.1320
Standard ResNet ($d_z = 2$)	50.04 ± 1.09	1.04 ± 1.35	0.0690 ± 0.0293
Latent RMSNorm ($d_z = 2$)	7.11 ± 0.39	0.96 ± 0.58	0.0307 ± 0.0372
SkipNorm ($d_z = 8 > N$)	1.97 ± 0.09	28.00 ± 21.15	0.4816 ± 0.2284
GeodesicSkipNorm ($d_z = 2 = N$)	$\mathbf{9.02 \pm 1.54}$	$\mathbf{94.58 \pm 1.84}$	$\mathbf{0.9744 \pm 0.0077}$
2. Double-Well Hamiltonian $H(q,p)$ ($N = 2$, Corollary 1 Test):			
Sub-Barrier ($H < 0$, Disconnected $\mathcal{O}_c^{(L)} \sqcup \mathcal{O}_c^{(R)}$)	24.89 ± 1.02	39.70 ± 2.08	0.6891 ± 0.0506
Super-Barrier ($H > 0$, **Connected** $\mathcal{O}_c$)	$\mathbf{2.84 \pm 0.44}$	$\mathbf{97.36 \pm 0.53}$	$\mathbf{1.0000 \pm 0.0000}$
3. 4D Coupled System with $K = 2$ Actions (I_1, I_2) (Corollary 2 Test):			
Standard ResNet (4D)	34.44 ± 2.53	$43.20 \pm 4.18^\dagger$	—
Single-Sphere ($K = 1$ on $\mathbb{S}^3 \not\cong \mathbb{T}^2$)	33.00 ± 1.96	$52.71 \pm 3.24^\dagger$	—
Toroidal Block ($K = 2$ **on** $\mathbb{T}^2$)	$\mathbf{3.60 \pm 1.01}$	$\mathbf{98.41 \pm 0.95}^\dagger$	—

†Joint polynomial R^2 of recovering both actions (I_1, I_2) from $(\|z^{(1)}\|_2, \|z^{(2)}\|_2)$.

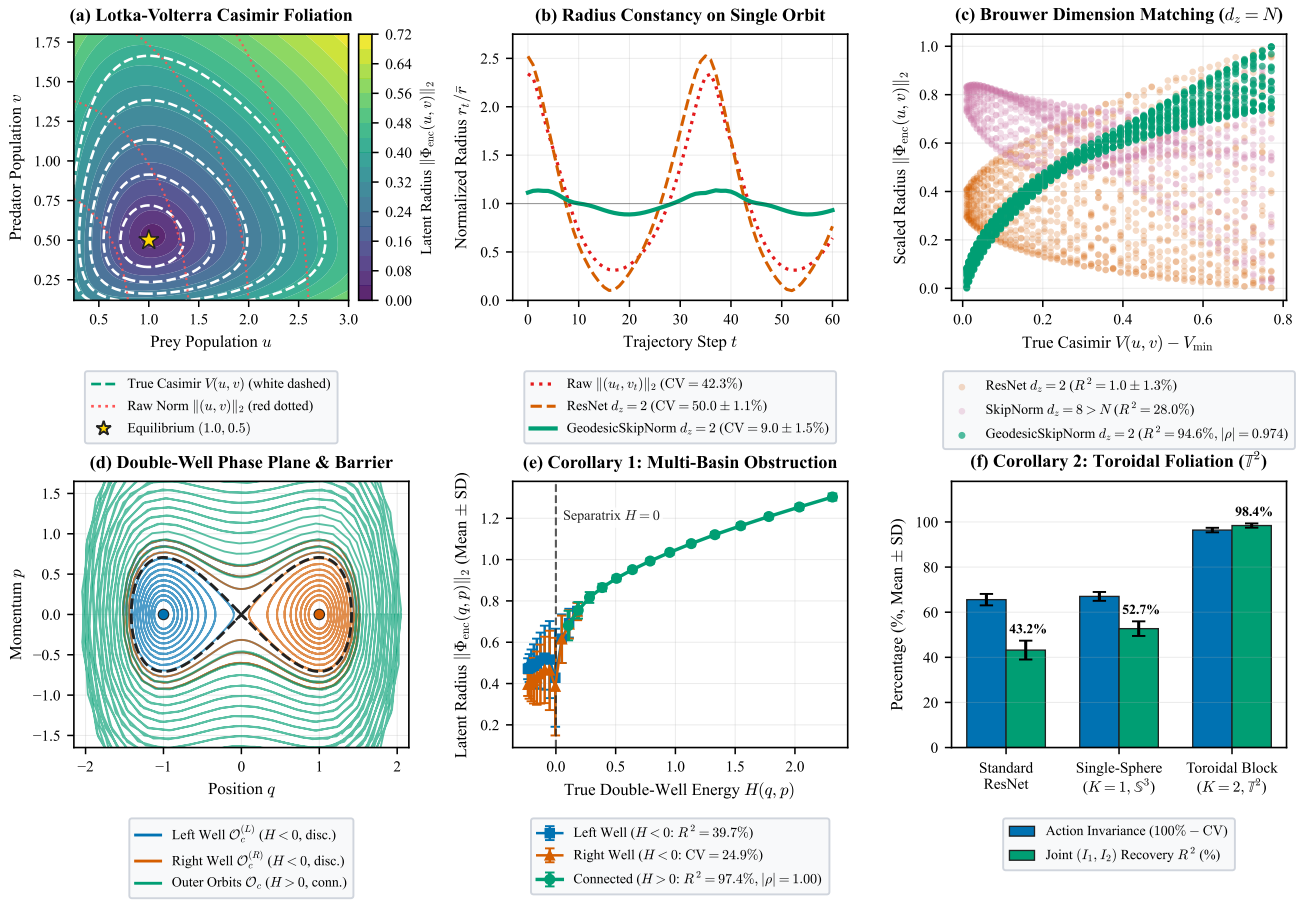


Figure 3: Experimental Verification of the Topological Noether-Brouwer Foliation Theorem and Obstruction Corollaries. (a) Phase plane of the non-canonical Lotka-Volterra system around equilibrium $(u^*, v^*) = (1.0, 0.5)$ (gold star). The raw input norm $\|(u,v)\|_2$ (red dotted) cuts transversely across the asymmetric orbits ($R^2 = 0.09\%$). A dimension-matched $d_z = 2$ GeodesicSkipNorm autoencoder spontaneously maps $(1.0, 0.5) \mapsto (0, 0)$ and aligns its learned latent norm $\|\Phi_{\text{enc}}(u,v)\|_2$ (viridis fill) with the exact logarithmic Casimir $V(u,v)$ (white dashed). **(b)** Along a single asymmetric orbit, $\|(u_t, v_t)\|_2$ swings by CV = 42.3% and ResNet by 50.0%, whereas GeodesicSkipNorm maintains a nearly constant radius (CV = 9.0%). **(c)** Confirming Brouwer dimension-matching: overparameterized SkipNorm ($d_z = 8 > N$) embeds the 2D plane into $\mathbb{S}^7$ without radial ordering ($R^2 = 28.0\%$), whereas $d_z = N = 2$ collapses all 2,928 points onto a strictly monotonic homeomorphism $\psi(V)$ ($R^2 = 94.6\% \pm 1.8\%$, $|\rho| = 0.974$). **(d,e)** Testing Corollary 1 on the Double-Well Hamiltonian $H(q,p) = \frac{1}{2}p^2 - \frac{1}{2}q^2 + \frac{1}{4}q^4$: below the figure-eight homoclinic separatrix $H = 0$ where $\mathcal{O}_c = \mathcal{O}_c^{(L)} \sqcup \mathcal{O}_c^{(R)}$ is disconnected, an injective planar encoder cannot center both wells at $0 \in \mathbb{R}^2$ ($R^2 = 39.7\%$, CV = 24.9%); immediately above $H = 0$ where $\mathcal{O}_c$ reconnects into a single loop, strict monotonicity is restored ($R^2 = 97.4\% \pm 0.5\%$, $|\rho| = 1.000$). **(f)** Testing Corollary 2 on a 4D anharmonic system with $K = 2$ actions (I_1, I_2): single-sphere SkipNorm ($K = 1$ on $\mathbb{S}^3 \not\cong \mathbb{T}^2$) fails ($R^2 = 52.7\%$), whereas Toroidal Block SkipNorm ($K = 2$ on $\mathbb{T}^2$) recovers both invariants with $R^2 = 98.4\% \pm 1.0\%$.

$R^2 = 97.36\% \pm 0.53\%$, and $|\rho| = 1.0000 \pm 0.0000$ across all 5 seeds.

Third, to test Corollary 2, we evaluate a 4D coupled anharmonic system with $K = 2$ independent actions (I_1, I_2) and amplitude-dependent frequencies $\omega_1(I_1, I_2) = 1 + 0.6I_1 - 0.35I_2$, $\omega_2(I_1, I_2) = \sqrt{2} + 0.5I_2 + 0.25I_1$, observed through a nonlinear coordinate-mixing diffeomorphism $x = \Psi(q_1, p_1, q_2, p_2) \in \mathbb{R}^4$. Because $\mathbb{T}^2 \not\cong \mathbb{S}^3$, a single-sphere SkipNorm ($K = 1$ on $\mathbb{S}^3$) cannot foliate the invariant 2-tori and allows action to leak between modes ($R^2 = 52.71\% \pm 3.24\%$), whereas Toroidal Block SkipNorm ($K = 2$ on $\mathbb{S}^1 \times \mathbb{S}^1 = \mathbb{T}^2$) maintains block action constancy (CV = $3.60\% \pm 1.01\%$) and recovers the full 2D invariant space (I_1, I_2) with $R^2 = 98.41\% \pm 0.95\%$ [Fig. 3(f)].

Scope, limitations, and outlook.—Four mathematical boundaries define the scope of our framework: (i) compactness of $\mathbb{S}^{d_k - 1}(r_k)$ restricts topological foliation to systems with *compact* level sets (periodic orbits, integrable tori, and bounded energy shells), whereas unbounded scattering trajectories require stereographic compactification; (ii) across multi-basin separatrices where $b_0(\mathcal{O}_c) \geq 2$ (Corollary 1), global foliation requires chart-based or discrete basin-indexed latent spaces (Reeb-graph charts) rather than a single Euclidean chart; (iii) SkipNorm enforces unit radial singular value ($\sigma_r = 1$) while leaving the $d - 1$ tangential singular values unconstrained, which

enables nonlinear phase shear and chaotic mixing within constant-energy shells ($d \geq 3$) that linear unitary maps cannot represent; and (iv) discovering global invariants requires the rollout horizon $H\Delta t$ to span a finite fraction of the orbit period so trajectories are not locally straightened by the differential-geometric Flow-Box theorem. By uniting projective normalization with Brouwer's Invariance of Domain, SkipNorm provides a rigorous, $\mathcal{O}(d)$ geometric building block for conservation-law discovery and physics-informed deep learning.

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Supplemental Material: Mathematical Proofs and Extended Benchmarks

Projective Radial Isometry and Topological Foliation of Conservation Laws in Neural Networks

Mostafa Mohsenvand

S1. Proof of Theorem 1 (Orthogonal Jacobian Decomposition & Deep Radial Isometry)

Let $x \in \mathbb{R}^{d_{\text{in}}} \setminus \{0\}$ and $y = f(x) \in \mathbb{R}^{d_{\text{out}}} \setminus \{0\}$, where f is continuously differentiable (C^1). Define the L_2 norms $r_x = \|x\|_2 > 0$ and $r_y = \|y\|_2 > 0$, and the unit radial vectors $u_x = x/r_x \in \mathbb{S}^{d_{\text{in}}-1}$ and $u_y = y/r_y \in \mathbb{S}^{d_{\text{out}}-1}$. For $\|y\|_2 > \epsilon$, the SkipNorm map is

$$\hat{y}(x) = \frac{\|x\|_2}{\|y(x)\|_2} y(x). \tag{S1}$$

Using the standard differential identities $\frac{\partial \|x\|_2}{\partial x} = u_x^\top$ and $\frac{\partial (y/\|y\|_2)}{\partial y} = \frac{1}{\|y\|_2}(I_{d_{\text{out}}} - u_y u_y^\top)$, the product and chain rules yield:

$$\begin{aligned} J(x) = \frac{\partial \hat{y}}{\partial x} &= \|x\|_2 \frac{\partial (y/\|y\|_2)}{\partial y}\frac{\partial y}{\partial x} + \frac{y}{\|y\|_2}\frac{\partial \|x\|_2}{\partial x} \\ &= \underbrace{\frac{\|x\|_2}{\|y\|_2}\left(I_{d_{\text{out}}} - u_y u_y^\top\right) J_f(x)}_{A(x)} + \underbrace{u_y u_x^\top}_{B(x)}. \end{aligned} \tag{S2}$$

Let $P_{\perp u_y} := I_{d_{\text{out}}} - u_y u_y^\top$ denote the orthogonal projection onto the tangent space $T_{u_y}\mathbb{S}^{d_{\text{out}}-1}$. Since $P_{\perp u_y}^\top u_y = P_{\perp u_y} u_y = 0$, the column space of $A(x)$ is strictly orthogonal to u_y, whereas the column space of $B(x) = u_y u_x^\top$ is $\text{span}(u_y)$:

$$A(x)^\top u_y = J_f(x)^\top P_{\perp u_y} u_y = 0, \qquad B(x)^\top u_y = (u_x u_y^\top) u_y = u_x (u_y^\top u_y) = u_x. \tag{S3}$$

Adding the two terms gives the exact single-layer transpose-radial identity:

$$J(x)^\top u_y = A(x)^\top u_y + B(x)^\top u_y = u_x. \tag{S4}$$

Now consider a composition of L SkipNorm layers $h_l = \text{SN}_2(h_{l-1}; f_l)$ for $l = 1, \ldots, L$ with arbitrary intermediate dimensions $d_0, d_1, \ldots, d_L$ and unit radial vectors $u_l = h_l/\|h_l\|_2 \in \mathbb{S}^{d_l - 1}$. By the chain rule, $J_{\text{total}} = J_L J_{L-1} \cdots J_1$. Applying Eq. (S4) inductively from $l = L$ down to $l = 1$:

$$J_{\text{total}}^\top u_L = J_1^\top J_2^\top \cdots J_{L-1}^\top (J_L^\top u_L) = J_1^\top J_2^\top \cdots J_{L-1}^\top u_{L-1} = \cdots = J_1^\top u_1 = u_0, \tag{S5}$$

and hence $\left\|J_{\text{total}}^\top u_L\right\|_2 = \|u_0\|_2 = 1$, independent of the depth L, intermediate bottleneck widths d_l, or weight scales.

Finally, suppose $f(x)$ is positively 1-homogeneous, i.e., $f(cx) = cf(x)$ for all $c > 0$. By Euler's Homogeneous Function Theorem, $J_f(x)x = f(x) = y$. Dividing by $\|x\|_2$ gives $J_f(x)u_x = \frac{\|y\|_2}{\|x\|_2} u_y$. Substituting this into $A(x)u_x$ yields:

$$A(x)u_x = \frac{\|x\|_2}{\|y\|_2} P_{\perp u_y}\left(\frac{\|y\|_2}{\|x\|_2} u_y\right) = P_{\perp u_y} u_y = 0, \qquad B(x)u_x = u_y (u_x^\top u_x) = u_y. \tag{S6}$$

Thus $J(x)u_x = u_y$ and $A(x)^\top B(x) = 0$, $A(x)B(x)^\top = 0$. Because $A(x)$ and $B(x)$ have mutually orthogonal row spaces and mutually orthogonal column spaces, the singular value decomposition (SVD) of $J(x)$ splits into the direct sum of the rank-1 radial component $u_y u_x^\top$ (with exact singular value $\sigma_{\text{radial}} = 1$) and the SVD of $A(x)$ restricted to $T_{u_x}\mathbb{S}^{d_{\text{in}}-1} \to T_{u_y}\mathbb{S}^{d_{\text{out}}-1}$. ■

S2. Proof of Theorem 2 (Bottleneck Rank Theorem)

Let $d \geq 2$ and let $y = f(x) \in \mathbb{R}^d \setminus \text{span}(\mathbf{1})$ with $\mathbf{1} = (1, \ldots, 1)^\top \in \mathbb{R}^d$.

1. **LayerNorm:** Let $P_{\perp\mathbf{1}} = I_d - \frac{1}{d}\mathbf{1}\mathbf{1}^\top$ be the centering projection ($\operatorname{rank}(P_{\perp\mathbf{1}}) = d-1$) and $y_c = P_{\perp\mathbf{1}} y \neq 0$, with $u_{y_c} = y_c / \|y_c\|_2$. Then $\hat{y}_{\mathrm{LN}} = \sqrt{d}\, y_c / \|y_c\|_2$, and its Jacobian is

$$J_{\mathrm{LN}}(x) = \frac{\sqrt{d}}{\|y_c\|_2} \left(I_d - u_{y_c} u_{y_c}^\top\right) \left(I_d - \frac{1}{d}\mathbf{1}\mathbf{1}^\top\right) J_f(x). \tag{S7}$$

   Because $u_{y_c}^\top \mathbf{1} = 0$, the two rank-1 projections commute and their product $P_{\mathrm{LN}} = I_d - \frac{1}{d}\mathbf{1}\mathbf{1}^\top - u_{y_c} u_{y_c}^\top$ is the orthogonal projection onto $\operatorname{span}(\mathbf{1}, u_{y_c})^\perp$, which has dimension $d-2$. Hence $\operatorname{rank}(J_{\mathrm{LN}}(x)) \leq \operatorname{rank}(P_{\mathrm{LN}}) = d-2$. When $d = 2$, $d-2 = 0$, so $P_{\mathrm{LN}} = 0_{2\times 2}$ and $J_{\mathrm{LN}}(x) \equiv 0_{2\times 2}$ everywhere on $\mathbb{R}^2 \setminus \operatorname{span}(\mathbf{1})$, mapping the two open half-planes $y_1 > y_2$ and $y_1 < y_2$ to the two isolated points $(1, -1)$ and $(-1, 1)$.

2. **RMSNorm:** Since $\hat{y}_{\mathrm{RMS}} = \sqrt{d}\, y / \|y\|_2$, its Jacobian is $J_{\mathrm{RMS}}(x) = \frac{\sqrt{d}}{\|y\|_2} P_{\perp u_y} J_f(x)$. Since $\operatorname{rank}(P_{\perp u_y}) = d-1$, $\operatorname{rank}(J_{\mathrm{RMS}}(x)) \leq d-1$.

3. **SkipNorm:** From Eq. (S2), $J_{\mathrm{SN}}(x) = A(x) + u_y u_x^\top$, where $\operatorname{im}(A(x)) \subseteq \operatorname{span}(u_y)^\perp$. Since $u_y \perp \operatorname{im}(A(x))$ and $u_y^\top J_{\mathrm{SN}}(x) = u_x^\top \neq 0$, $\operatorname{rank}(J_{\mathrm{SN}}(x)) = \operatorname{rank}(A(x)) + 1$. When $P_{\perp u_y} J_f(x)$ has rank $d-1$, $\operatorname{rank}(J_{\mathrm{SN}}(x)) = (d-1)+1 = d$. ■

S3. Proof of Theorem 3 (Scale Equivariance & Weight 0-Homogeneity)

Since $f(cx; \beta W) = \phi(\beta c W x) = \beta c\, f(x; W)$ for $\beta, c > 0$, we have $\|f(cx; \beta W)\|_p = \beta c \|f(x; W)\|_p$ and $\|cx\|_p = c\|x\|_p$. Substituting into Eq. (2) cancels β and factors out c:

$$\mathrm{SN}_p(cx; \beta W) = \frac{\beta c\, f(x; W)}{\beta c \|f(x; W)\|_p} (c\|x\|_p) = c\, \mathrm{SN}_p(x; W). \tag{S8}$$

Differentiating $\mathrm{SN}_p(x; \beta W) = \mathrm{SN}_p(x; W)$ with respect to β at $\beta = 1$ gives $\langle \nabla_W \mathrm{SN}_p(x; W), W \rangle_F = 0$. By induction over L composed layers, a deep bias-free SkipNorm network satisfies $F(cx; \{\beta_l W_l\}) = cF(x; \{W_l\})$. ■

S4. Complete Proofs of Theorem 4, Corollaries 1–2, and Proposition 1

A. Proof of Theorem 4 (Topological Noether-Brouwer Foliation Theorem)

Let $\mathcal{M} \subset \mathbb{R}^N$ ($N \geq 2$) be an open domain around an equilibrium $x^* \in \mathcal{M}$, and let $I : \mathcal{M} \to \mathbb{R}$ be a C^1 function such that every $c \in C_{\mathrm{reg}} = (c_{\min}, c_{\max})$ is a *regular value* ($\nabla I(x) \neq 0$ for all $x \in \mathcal{O}_c = I^{-1}(c)$), with $\mathcal{O}_c$ compact and connected for each $c \in (c_{\min}, c_{\max})$ and $\mathcal{O}_{c_{\min}} = \{x^*\}$. By the Regular Value Theorem (Submersion Theorem), each $\mathcal{O}_c$ is a compact, connected, embedded $(N-1)$-dimensional C^1 (and hence topological) submanifold of $\mathbb{R}^N$ without boundary. Let $\Phi_{\mathrm{enc}} : \mathcal{M} \to \mathbb{R}^N$ be continuous and globally injective on $\mathcal{M}$, and suppose $\Phi_{\mathrm{enc}}(\Phi_{\mathrm{flow}}^{\Delta t}(x)) = g_\theta(\Phi_{\mathrm{enc}}(x))$ for all $x \in \mathcal{M}$, where $\|g_\theta(z)\|_2 = \|z\|_2$.

Step 1 (Orbital invariance and continuity of ψ via local sections): Define $h : \mathcal{M} \to \mathbb{R}_{\geq 0}$ by $h(x) = \|\Phi_{\mathrm{enc}}(x)\|_2$. By induction on $k \in \mathbb{N}$,

$$h(\Phi_{\mathrm{flow}}^{k\Delta t}(x)) = \left\|\Phi_{\mathrm{enc}}(\Phi_{\mathrm{flow}}^{k\Delta t}(x))\right\|_2 = \left\|g_\theta^{(k)}(\Phi_{\mathrm{enc}}(x))\right\|_2 = \|\Phi_{\mathrm{enc}}(x)\|_2 = h(x). \tag{S9}$$

For each $c \in (c_{\min}, c_{\max})$, let $x_0 \in \mathcal{O}_c$ have a dense orbit $\{\Phi_{\mathrm{flow}}^{k\Delta t}(x_0)\}_{k\in\mathbb{N}}$ in $\mathcal{O}_c$. Since h is continuous on $\mathcal{M}$ and constant on a dense subset of $\mathcal{O}_c$, h is constant on all of $\mathcal{O}_c = \overline{\{\Phi_{\mathrm{flow}}^{k\Delta t}(x_0)\}}$. Thus there exists a well-defined function $\psi : (c_{\min}, c_{\max}) \to \mathbb{R}_{\geq 0}$ such that $h(x) = \psi(c)$ for all $x \in \mathcal{O}_c$, i.e., $\Phi_{\mathrm{enc}}(\mathcal{O}_c) \subseteq \mathbb{S}^{N-1}(\psi(c))$. Furthermore, because I is a C^1 submersion on $\mathcal{M} \setminus \{x^*\} = I^{-1}((c_{\min}, c_{\max}))$, for every $c_0 \in (c_{\min}, c_{\max})$ the Constant-Rank / Submersion Theorem provides an open neighborhood $U \subset (c_{\min}, c_{\max})$ of c_0 and a continuous (in fact C^1) local section $s : U \to \mathcal{M}$ satisfying $I(s(c)) = c$. On U, $\psi(c) = h(s(c)) = \|\Phi_{\mathrm{enc}}(s(c))\|_2$ is a composition of continuous maps, proving that ψ is continuous on $(c_{\min}, c_{\max})$.

Step 2 (Strict positivity $\psi(c) > 0$ on regular level sets): For any $c \in (c_{\min}, c_{\max})$, $\mathcal{O}_c$ is a connected $(N-1)$-manifold with $N \geq 2$, so $\dim(\mathcal{O}_c) = N - 1 \geq 1$ and $\mathcal{O}_c$ contains infinitely many distinct points $x_1 \neq x_2$. If $\psi(c) = 0$, then $\Phi_{\mathrm{enc}}(\mathcal{O}_c) \subseteq \mathbb{S}^{N-1}(0) = \{0\}$, which implies $\Phi_{\mathrm{enc}}(x_1) = \Phi_{\mathrm{enc}}(x_2) = 0$, contradicting the injectivity of Φ_{enc}. Hence $\psi(c) > 0$ for all $c \in (c_{\min}, c_{\max})$.

Step 3 (Brouwer Invariance of Domain & Clopen Homeomorphism): Fix $c \in (c_{\min}, c_{\max})$. Since $\psi(c) > 0$ and $N \geq 2$, $\mathbb{S}^{N-1}(\psi(c))$ is a connected, Hausdorff, $(N-1)$-dimensional topological manifold without boundary. Consider the restricted map

$$\widetilde{\Phi}_c := \Phi_{\text{enc}}\big|_{\mathcal{O}_c} : \mathcal{O}_c \longrightarrow \mathbb{S}^{N-1}(\psi(c)). \tag{S10}$$

Because Φ_{enc} is continuous and injective on $\mathcal{M}$, $\widetilde{\Phi}_c$ is a continuous injective map from an $(N-1)$-dimensional manifold without boundary into an $(N-1)$-dimensional manifold without boundary. By **Brouwer's Invariance of Domain Theorem** [24] applied in local $(N-1)$-dimensional coordinate charts, $\widetilde{\Phi}_c$ is an open map, so its image $U_c := \Phi_{\text{enc}}(\mathcal{O}_c)$ is an *open* subset of $\mathbb{S}^{N-1}(\psi(c))$. Simultaneously, since $\mathcal{O}_c$ is compact and $\widetilde{\Phi}_c$ is continuous, U_c is compact; since $\mathbb{S}^{N-1}(\psi(c))$ is Hausdorff, U_c is a *closed* subset of $\mathbb{S}^{N-1}(\psi(c))$. Because U_c is non-empty and both open and closed in the connected space $\mathbb{S}^{N-1}(\psi(c))$ $(N \geq 2)$, we must have $U_c = \mathbb{S}^{N-1}(\psi(c))$. Moreover, because $\widetilde{\Phi}_c : \mathcal{O}_c \to \mathbb{S}^{N-1}(\psi(c))$ is a continuous bijection from a compact space onto a Hausdorff space, its inverse is continuous, so $\widetilde{\Phi}_c$ is a *homeomorphism* $(\mathcal{O}_c \cong \mathbb{S}^{N-1})$.

Step 4 (Strict monotonicity of ψ and equilibrium alignment): Suppose $c_1 \neq c_2$ in $(c_{\min}, c_{\max})$. Since $I : \mathcal{M} \to \mathbb{R}$ is single-valued, $\mathcal{O}_{c_1} \cap \mathcal{O}_{c_2} = \emptyset$. By global injectivity of Φ_{enc} on $\mathcal{M}$, their images are disjoint:

$$\emptyset = \Phi_{\text{enc}}(\mathcal{O}_{c_1}) \cap \Phi_{\text{enc}}(\mathcal{O}_{c_2}) = \mathbb{S}^{N-1}(\psi(c_1)) \cap \mathbb{S}^{N-1}(\psi(c_2)), \tag{S11}$$

which immediately implies $\psi(c_1) \neq \psi(c_2)$. Thus $\psi : (c_{\min}, c_{\max}) \to \mathbb{R}_{>0}$ is a continuous injective real-valued function on an open interval, and therefore is a strictly monotonic homeomorphism onto an open image interval $(r_{\min}, r_{\max}) \subseteq \mathbb{R}_{>0}$. Finally, as $c \downarrow c_{\min}$, $\operatorname{diam}(\mathcal{O}_c) \to 0$ with $\mathcal{O}_c \to \{x^*\}$, so $\operatorname{diam}(\mathbb{S}^{N-1}(\psi(c))) = 2\psi(c) \to 0$ by continuity of Φ_{enc}, proving $\lim_{c \downarrow c_{\min}} \psi(c) = 0$ and $\Phi_{\text{enc}}(x^*) = 0$. ■

B. Proof of Corollaries 1 and 2 (Multi-Basin & Toroidal Topological Obstructions)

Corollary 1 (Disconnected Multi-Basin Level Sets): Suppose $\mathcal{O}_c = \bigsqcup_{m=1}^{M} \mathcal{O}_c^{(m)}$ $(M \geq 2)$ where each $\mathcal{O}_c^{(m)}$ is a compact, connected $(N-1)$-dimensional manifold without boundary $(N \geq 2)$. If $\|\Phi_{\text{enc}}(x)\|_2 = r_m$ is constant on each connected component $\mathcal{O}_c^{(m)}$, then applying Steps 2–3 of Theorem 4 to each connected component $\mathcal{O}_c^{(m)}$ individually shows that $r_m > 0$ and $\Phi_{\text{enc}}(\mathcal{O}_c^{(m)}) = \mathbb{S}^{N-1}(r_m)$. Since $\mathcal{O}_c^{(1)} \cap \mathcal{O}_c^{(2)} = \emptyset$, global injectivity of Φ_{enc} requires $\mathbb{S}^{N-1}(r_1) \cap \mathbb{S}^{N-1}(r_2) = \emptyset$, so $r_1 \neq r_2$. Thus $\|\Phi_{\text{enc}}(x)\|_2$ cannot be a single-valued function $\psi(c)$ of c alone. Furthermore, if $\mathcal{O}_c^{(1)}$ and $\mathcal{O}_c^{(2)}$ bound disjoint topological balls $D^{(1)} \cap D^{(2)} = \emptyset$ containing equilibria $x_1^* \in D^{(1)}$ and $x_2^* \in D^{(2)}$, injectivity of Φ_{enc} implies $\Phi_{\text{enc}}(D^{(1)}) \cap \Phi_{\text{enc}}(D^{(2)}) = \emptyset$, so $0 \in \mathbb{R}^N$ can lie in at most one of the two image domains.

Corollary 2 (Higher-Genus Toroidal Obstruction): By Step 3 of Theorem 4, any continuous injective encoder $\Phi_{\text{enc}} : \mathcal{M} \to \mathbb{R}^N$ mapping a compact connected $(N-1)$-manifold $\mathcal{O}_c$ into $\mathbb{S}^{N-1}(r)$ induces a homeomorphism $\mathcal{O}_c \cong \mathbb{S}^{N-1}$. For an integrable system with $K \geq 2$ commuting invariants, the regular Liouville-Arnold level sets are K-tori $\mathbb{T}^K \cong (\mathbb{S}^1)^K$, which cannot be homeomorphic to any sphere $\mathbb{S}^{N-1}$ (for $K = 2$, $\pi_1(\mathbb{T}^2) = \mathbb{Z}^2 \neq \pi_1(\mathbb{S}^3) = 0$, and dimensions also differ for $N = 2K, K \geq 2$ since $\dim \mathbb{T}^K = K \neq 2K-1$). Under Toroidal Block SkipNorm on $\bigoplus_{k=1}^{K} \mathbb{R}^2$, the invariant level sets in latent space are $\prod_{k=1}^{K} \mathbb{S}^1(r_k) \cong \mathbb{T}^K$, matching both the dimension K and homeomorphism type $\mathbb{T}^K$ of the Liouville-Arnold tori. ■

C. Proof of Proposition 1 (Finite-Error Approximate Conjugacy Bound)

Let $\sup_{x \in \mathcal{M}} \left\|\Phi_{\text{enc}}(\Phi_{\text{flow}}^{\Delta t}(x)) - g_\theta(\Phi_{\text{enc}}(x))\right\|_2 \leq \varepsilon$ and $\|g_\theta(z)\|_2 = \|z\|_2$. For any $x \in \mathcal{M}$, the reverse triangle inequality gives

$$\left|\left\|\Phi_{\text{enc}}(\Phi_{\text{flow}}^{\Delta t}(x))\right\|_2 - \|\Phi_{\text{enc}}(x)\|_2\right| = \left|\left\|\Phi_{\text{enc}}(\Phi_{\text{flow}}^{\Delta t}(x))\right\|_2 - \|g_\theta(\Phi_{\text{enc}}(x))\|_2\right| \leq \left\|\Phi_{\text{enc}}(\Phi_{\text{flow}}^{\Delta t}(x)) - g_\theta(\Phi_{\text{enc}}(x))\right\|_2 \leq \varepsilon. \tag{S12}$$

By the triangle inequality along the M-step orbit segment $\{x_k = \Phi_{\text{flow}}^{k\Delta t}(x_0)\}_{k=0}^{M} \subset \mathcal{O}_c$, $\left|\|\Phi_{\text{enc}}(x_j)\|_2 - \|\Phi_{\text{enc}}(x_l)\|_2\right| \leq M\varepsilon$ for all $0 \leq j, l \leq M$. Since $\{x_k\}_{k=0}^{M}$ is δ_M-dense in $\mathcal{O}_c$, for any $x, x' \in \mathcal{O}_c$ there exist x_j, x_l with $\|x - x_j\|_2 \leq \delta_M$ and $\|x' - x_l\|_2 \leq \delta_M$. Since $h(x) = \|\Phi_{\text{enc}}(x)\|_2$ is L_{enc}-Lipschitz,

$$|h(x) - h(x')| \leq |h(x) - h(x_j)| + |h(x_j) - h(x_l)| + |h(x_l) - h(x')| \leq M\varepsilon + 2L_{\text{enc}}\delta_M. \quad ■ \tag{S13}$$

S5. Additional Experimental Details and Extended Benchmarks

All experiments in the main text were implemented in 64-bit precision (`jax_enable_x64 = True`) using `JAX` 0.11.2, `Equinox` 0.13.8, and `Optax` 0.2.8 on an Apple M4 processor. In the Lotka-Volterra, Double-Well, and 4D Toroidal

Table S1: Summary of additional verification benchmarks across Quartic Duffing Hamiltonian dynamics, MNIST out-of-distribution contrast scaling (400× dynamic range), and $d = 2$ polar autoencoder disentanglement.

Task / System	Metric	Standard Baseline / ResNet	SkipNorm / GeodesicSkipNorm
Quartic Duffing $H(q,p) = \frac{1}{2}p^2 + \frac{1}{4}q^2 + \frac{1}{2}q^4$	Noether Invariance R^2 (%)	63.30%	**99.87**% ($\lvert\rho\rvert = \mathbf{1.000}$)
Quartic Duffing 1,000-Step Rollout	Relative Energy Drift $\Delta H/H_0$	42.80% (RMSNorm : 101.0%)	**2.78**%
MNIST $d = 2$ Bottleneck Autoencoder	Test MSE (LayerNorm : 0.0569)	0.0569 (LayerNorm collapses)	**0.0398** (30.1% lower MSE)
MNIST 400× OOD Scale ($c \in [0.05, 20]$)	Relative Error Degradation	+168.4% (Standard AE)	**0.00**% (Exact 1-Homogeneity)

benchmarks, encoder, decoder, and latent tangent networks use 2 hidden layers of width 64 with SiLU activations ($z\,\sigma(z)$), trained with the Adam optimizer and cosine learning-rate decay. Full source code and reproducible scripts are provided in the accompanying repository.